Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/17/2025
PI Name:	Blaber, Elizabeth Ph.D.		
Project Title:	The Role of the Hematopoietic Stem Cell in the Hepatic Response to Simulated Spaceflight		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Organization Name:	Rensselaer Polytechnic Institute		
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City:	Troy	**State:**	NY
Zip Code:	12180	**Congressional District:**	20
Comments:	NOTE: Moved to Rensselaer Polytechnic Institute in fall 2019 from NASA Ames Research Center.		
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	08/01/2020	**End Date:**	07/31/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2026 per F. Hernandez/ARC (Ed., 9/17/25) NOTE: End date changed to 7/31/2025 per F. Hernandez/ARC (Ed., 8/29/24) NOTE: End date changed to 7/31/2024 per NSSC information (Ed., 11/13/23) NOTE: End date changed to 7/31/2023 per NSSC information (Ed., 8/24/22) NOTE: End date changed to 7/31/2022 per F. Hernandez/ARC (Ed., 7/27/21)		
Key Personnel Changes/Previous PI:	NOTE: CoPI Karen Jonscher has been at University of Oklahoma Health Sciences Center since 2019; affiliation changed in this 2021 record.		
COI Name (Institution):	Jonscher, Karen Ph.D. (CoPI: University of Oklahoma Health Sciences Center)		
Grant/Contract No.:	80NSSC20K0289		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts embarking on long duration spaceflight missions will be exposed to multiple and potentially synergistic risks, two of which are exposure to altered inertial forces (gravity) and long-term exposure to low-dose/low-dose-rate radiation. In our previous studies using spaceflight and simulated spaceflight (hindlimb unloading and/or space-like radiation), we identified increased oxidative stress and activation of the inflammatory response as major contributors to physiological changes associated with spaceflight. These histological changes include alterations to the cardiovascular system, the musculoskeletal system, metabolism, and the liver. In previous studies, we have found that spaceflight results in significant alterations to bone marrow stem cell populations, including down-regulation of the hematological system and cellular development. Additionally, an increase in populations of pro-inflammatory macrophages has been observed in multiple spaceflight experiments; we postulate this results from activation of inflammatory pathways in bone marrow populations during spaceflight exposure, leading to phenotypic shifts in hematopoietic stem cells and monocyte activation. Therefore, we aim to study mechanisms regulating activation of pro-inflammatory macrophages and their role in inducing liver injury, including steatosis and fibrotic scarring. Furthermore, we aim to assess the effectiveness of a dietary countermeasure available for over-the-counter human use, coenzyme pyrroloquinoline quinone (PQQ), on mitigating the observed activation of hepatic inflammation and the hypothesized activation of pro-inflammatory macrophages (and resulting liver damage) in response to simulated spaceflight conditions. This research will address Space Biology Research Plan's questions SBP CMB (Cell and Molecular Biology) 1-4 (CMB-1: What are the underlying genetic, molecular and biochemical mechanisms of the cell that are influenced by gravitational changes and the space environment?; CMB-2: What are the systems biology mechanisms and pathways for cell function that are responsive to gravitational changes and the spaceflight conditions?; CMB 3: What are the cellular and molecular bases for gravitational changes and space environment effects to tissue, organ, and whole organisms?; CMB 4: How does the cell respond to the biophysical changes (thermal, fluids, etc) that occur due to microgravity and how does this impact overall tissue function and changes?), CMB 6-8 (CMB 6: Does the space environment affect cellular and molecular functions in a manner that impact tissue morphogenesis or development?; CMB 7: What are the driving space environmental factors that are sensed and responded to at the cellular and molecular levels?; CMB 8: How does the space environment affect stem cell function and differentiation and how do the changes affect normal tissue function, regeneration, and embryogenesis), and AN (Animal Biology) 1-3 (AN-1: How does the space environment affect basic physiological functions and the applicable regulatory mechanisms and pathways, e.g., sensory-motor control, behavior, vision, metabolism, organ structure and function, aging and lifespan? Are there sex differences in the physiological responses?; AN-2: Do microgravity and ground-based microgravity analogs alter radiation damage and repair mechanisms in animals?; AN-3: How do animals sense and respond to alternations in gravity, i.e., what cellular and molecular mechanisms and pathways are used?).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research aims to understand the role of bone marrow derived cell populations on liver health under simulated microgravity conditions. Therefore, this research will provide important insight into how spaceflight affects both stem cell populations and peripheral tissues. Additionally, this project will test the effectiveness of an antioxidant countermeasure to combat the detrimental liver and bone marrow related changes during spaceflight.
Task Progress:	All Aims are complete for this project. Manuscript preparation is underway.
Bibliography Type:	Description: (Last Updated: 04/03/2025)